Manage Library Catalog

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<tr><td>Use Case Name:</td><td>Manage Library Catalog</td></tr>
<tr><td>Scenario:</td><td>Librarian maintains the catalog database, while Students and Faculty search, view, and select library resources.</td></tr>
<tr><td>Triggering Event:</td><td>Student/Faculty accesses the catalog search interface, or a Librarian accesses the catalog management dashboard.</td></tr>
<tr><td>Brief Description:</td><td>Encompasses catalog searching and viewing item availability by students/faculty (View Library Catalog), as well as administrative CRUD operations (add, edit, delete) performed by the Librarian. Interfaces with Koha ILS.</td></tr>
<tr><td>Actors:</td><td>Student/Faculty, Librarian, Koha ILS (External System)</td></tr>
<tr><td>Related Use Cases:</td><td>Includes: View Library Catalog. Extends: Manage Borrowed Books.</td></tr>
<tr><td>Stakeholders:</td><td>• Student/Faculty: To easily find and view required resources.
• Librarian: To maintain accurate library inventory records.</td></tr>
<tr><td>Preconditions:</td><td>• Student/Faculty: Successfully signs in to the LIRA system using their APC Microsoft 365 account.
• Librarian: Successfully signs in to the Koha system using staff credentials.
• Catalog database is online and accessible.</td></tr>
<tr><td>Postconditions:</td><td>• Catalog database is updated and saved in Koha ILS.
• Matching catalog items are accurately displayed to users.</td></tr>
<tr><td>Flow of Activities:</td><td>
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<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Student/Faculty signs in to LIRA and enters search criteria.</td><td>1.1 Retrieves matching database records and displays list.</td></tr>
<tr><td>2. Student/Faculty selects a specific item from search results.</td><td>2.1 Displays detailed information, availability, and location.</td></tr>
<tr><td>3. Librarian signs in to Koha and accesses catalog dashboard.</td><td>3.1 Displays the current catalog list.</td></tr>
<tr><td>4. Librarian</td><td>4.1 Validates data,</td></tr>
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	submits new item details, edits records, or confirms deletion.	updates catalog database, and displays success message.
Exception Conditions:	• If no items match search criteria, the system displays a 'No results found' message. • If data validation fails during an update (e.g., missing ISBN), the system highlights errors and prompts for correction. • If the Librarian modifies/deletes an item currently checked out, the system invokes the borrowed books process to resolve status conflicts. • If Koha authentication fails, Koha displays an error message and denies access.	

Manage Borrowed Books

Use Case Name:	Manage Borrowed Books	
Scenario:	Students and Faculty initiate borrow requests, while the Librarian manages circulation status and resolves lost records.	
Triggering Event:	User initiates a borrow request, or a Librarian processes a check-in, renewal, or flags a lost book.	
Brief Description:	Handles user borrowing requests (Borrow Books) and administrative circulation tracking. It records transactions by communicating with the Koha ILS.	
Actors:	Student/Faculty, Librarian, Koha ILS (External System)	
Related Use Cases:	Extension of: Manage Library Catalog. Includes: Borrow Books.	
Stakeholders:	• Student/Faculty: To successfully acquire books. • Librarian: To track inventory lifecycle and circulation.	
Preconditions:	• Student/Faculty: Successfully signs in to the LIRA system with a valid account. • Librarian: Successfully signs in to the Koha system using staff credentials.	
Postconditions:	• Item status is updated to 'Checked Out' or returned. • Transaction log is automatically committed to Koha ILS.	
Flow of Activities:	**Actor**	**System**

	1. Student/Faculty initiates borrow request and accepts terms.	1.1 Updates item status to 'Checked Out', displays receipt and due date.
	2. Librarian opens LIRA interface to view circulation.	2.1 Queries Koha and displays synchronized status of borrowed books.
	3. Librarian initiates a return or renewal.	3.1 Commits transaction to Koha, validates and updates local interface.
	4. Librarian flags a lost book.	4.1 Marks borrowed record as resolved/lost and removes from active circulation.
Exception Conditions:	• If the user has a blocked account or overdue items, the system denies the borrow request and displays a warning. • If the item becomes unavailable during the transaction, the system cancels the process and notifies the user. • If the Koha system is unreachable, the system queues the transaction and alerts the Librarian.	

Use Case Name:	Manage AI Chatbot
Scenario:	Students and Faculty ask AI questions for research assistance, while the Librarian configures the AI knowledge base.
Triggering Event:	User submits a query to the chat widget, or Librarian accesses the configuration panel.
Brief Description:	Encompasses automated response generation for student queries (Ask AI Chatbot) and administrative tuning of prompts and knowledge sources by the Librarian (Manage Chatbot).
Actors:	Student/Faculty, Librarian, AI Engine (OpenRouter)
Related Use Cases:	Includes: Ask AI Chatbot.
Stakeholders:	• Student/Faculty: For immediate research assistance.

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<tr><td></td><td>• Librarian: To audit AI interactions and ensure accuracy.</td></tr>
<tr><td>Preconditions:</td><td>• Student/Faculty: Successfully signs in to the LIRA system.
• Librarian: Successfully signs in to the LIRA system using administrative credentials.</td></tr>
<tr><td>Postconditions:</td><td>• Relevant response is generated and displayed.
• Chatbot configuration or knowledge sources are updated.</td></tr>
<tr><td>Flow of Activities:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Student/Faculty types and submits a question into the chat widget.</td><td>1.1 Routes the query, formats retrieved answer, and displays it in interface.</td></tr>
<tr><td>2. Librarian signs in with admin credentials and opens configuration panel.</td><td>2.1 Recognizes admin role, redirects to dashboard, displays current parameters and prompt logs.</td></tr>
<tr><td>3. Librarian uploads a new knowledge document.</td><td>3.1 Applies the new source to the AI engine.</td></tr>
<tr><td>4. Librarian edits configuration and saves changes.</td><td>4.1 Validates and applies updated settings.</td></tr>
<tr><td>5. Librarian deletes an obsolete document.</td><td>5.1 Removes the document so the AI ceases to use it for context.</td></tr>
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<tr><td>Exception Conditions:</td><td>• If the AI engine cannot understand the query, the system displays a fallback message asking the user to rephrase.
• If the AI service times out, the system apologizes and provides contact info for a human librarian.
• If the Librarian uploads an incompatible document, the system rejects it and shows an error message.
• If admin authentication fails, they are shown an error message and denied access.</td></tr>
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<tr><td>Use Case Name:</td><td>Manage Reports</td></tr>
<tr><td>Scenario:</td><td>The Librarian generates analytics regarding library usage, circulation data, and chatbot interactions.</td></tr>
<tr><td>Triggering Event:</td><td>Librarian requests a statistical or operational report from the dashboard.</td></tr>
<tr><td>Brief Description:</td><td>Allows the Librarian to compile, preview, adjust, and export data reports by extracting data from local LIRA databases and integrating circulation data from Koha.</td></tr>
<tr><td>Actors:</td><td>Librarian, Koha ILS (External System)</td></tr>
<tr><td>Related Use Cases:</td><td>N/A</td></tr>
<tr><td>Stakeholders:</td><td>• Library Management: To make data-driven decisions regarding system usage.</td></tr>
<tr><td>Preconditions:</td><td>• Librarian: Successfully signs in to Koha system using staff credentials.</td></tr>
<tr><td>Postconditions:</td><td>• Formatted report is generated, previewed, and available for download.</td></tr>
<tr><td>Flow of Activities:</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Librarian selects a report type and date range.</td><td>1.1 Retrieves data from LIRA and Koha, aggregates combined data.</td></tr>
<tr><td>2. Librarian views report preview on dashboard.</td><td>2.1 Displays report preview and provides downloadable file (PDF/Excel).</td></tr>
<tr><td>3. Librarian updates the date range or report type.</td><td>3.1 Re-aggregates data to reflect the new parameters.</td></tr>
<tr><td>4. Librarian dismisses or clears the report.</td><td>4.1 Removes the preview data from the dashboard.</td></tr>
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<tr><td>Exception Conditions:</td><td>• If Koha is offline when a report is generated, the system generates a partial report with a warning note.
• If the updated date range contains no data, the system outputs an empty report with a notification.</td></tr>
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